

Commerce XI Maths App

Exercise 2.1

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 2 - Measures of Dispersion | Total questions: 8

Question 1 - Enter answer · PG-26 · EX-Q1

Find range of the following data:

575, 609, 335, 280, 729, 544, 852, 427, 967, 250

Correct answer: 717

Explanation and steps

Use range = largest value - smallest value. Substitution gives 717.

Question 2 - Enter answer · PG-26 · EX-Q2

The following data gives number of typing mistakes done by Radha during a week.

Find the range of the data.

Day	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
No. of mistakes	15	20	21	12	17	10

Correct answer: 11

Explanation and steps

Use range = largest value - smallest value. Substitution gives 11.

Question 3 - Enter answer · PG-26 · EX-Q3

Find range for the following data.

Classes	62-64	64-66	66-68	68-70	70-72
Frequency	5	3	4	5	3

Correct answer: 10

Explanation and steps

Use range = largest value - smallest value. Substitution gives 10.

Question 4 - Enter answer · PG-26 · EX-Q4

Find Q.D. for the following data:

3, 16, 8, 15, 19, 11, 5, 17, 9, 5, 3.

Correct answer: 5.5

Explanation and steps

Locate Q1 and Q3, then use $Q.D. = (Q_3 - Q_1)/2$. The verified result is 5.5.

Question 5 - Enter answer · PG-26 · EX-Q5

Given below are the prices of shares of a company for the last 10 days.

172, 164, 188, 214, 190, 237, 200, 195, 208, 230

Find Q.D.

Correct answer: 17

Explanation and steps

Locate Q1 and Q3, then use $Q.D. = (Q_3 - Q_1)/2$. The verified result is 17.

Question 6 - Enter answer · PG-26 · EX-Q6

Calculate Q.D. for the following data.

X	24	25	26	27	28	29	30
F	6	5	3	2	4	7	3

Correct answer: 2

Explanation and steps

Locate Q1 and Q3, then use $Q.D. = (Q3 - Q1)/2$. The verified result is 2.

Question 7 - Enter answer · PG-26 · EX-Q7

Following data gives the age distribution of 250 employees of a firm.

Calculate Q.D. of the distribution.

Age (in years)	20-25	25-30	30-35	35-40	40-45	45-50
No. of employees	30	40	60	50	46	14

Correct answer: 5.625

Explanation and steps

Locate Q1 and Q3, then use $Q.D. = (Q3 - Q1)/2$. The verified result is 5.625.

Question 8 - Enter answer · PG-26 · EX-Q8

Following data gives the weight of boxes.

Calculate Q.D. for the data.

Weight (kg)	10-12	12-14	14-16	16-18	18-20	20-22
No. of boxes	3	7	16	14	18	2

Correct answer: 1.965

Explanation and steps

Locate Q1 and Q3, then use $Q.D. = (Q3 - Q1)/2$. The verified result is 1.965.